EXPERIMENT-3 UID

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Experiment 3 – File Renaming Using Different User Interfaces

Command Line Interface (CLI)

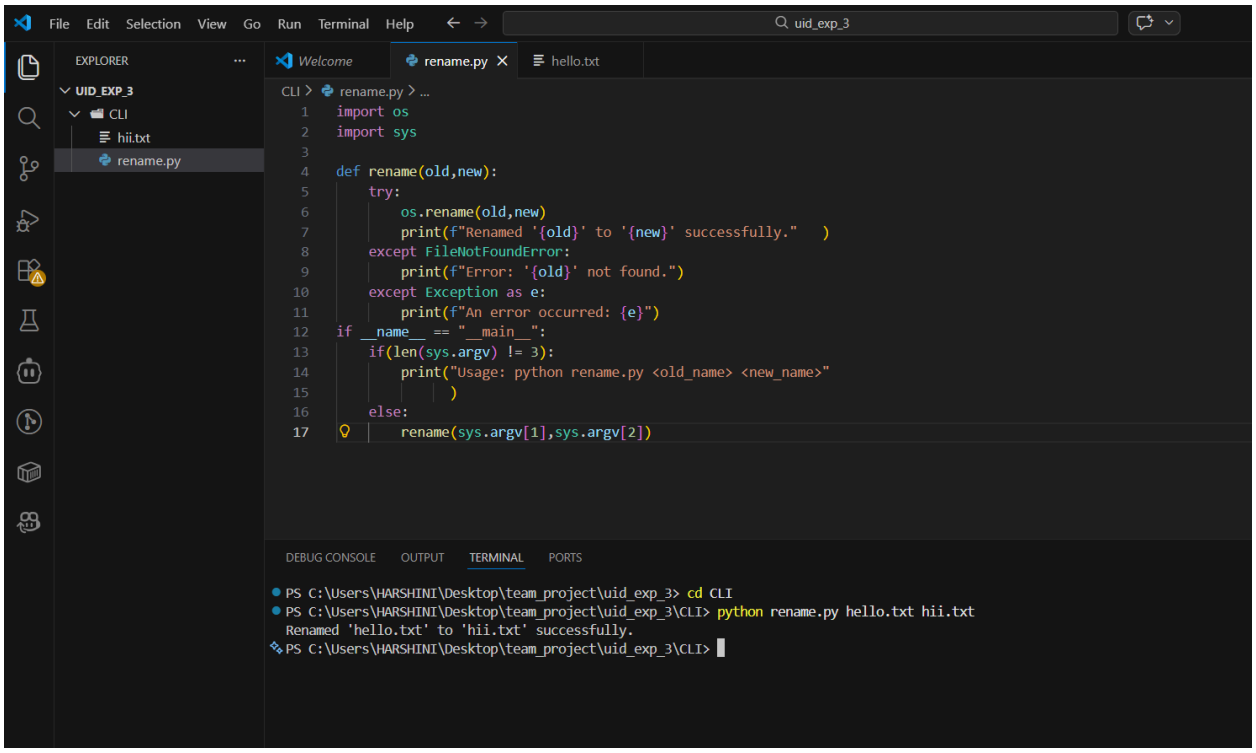
The CLI program allows users to rename files by entering commands in the terminal. The user provides the old filename and new filename as command-line arguments, and the program processes the input to rename the file. This method is fast and suitable for technical users.

Code:

```
import os

import sys



def rename(old,new):

    try:

        os.rename(old,new)

        print(f"Renamed '{old}' to '{new}' successfully."   )

    except FileNotFoundError:

        print(f"Error: '{old}' not found.")

    except Exception as e:

        print(f"An error occurred: {e}")

if __name__ == "__main__":
```

```
if(len(sys.argv) != 3):
    print("Usage: python rename.py <old_name> <new_name>"
        )
else:
    rename(sys.argv[1],sys.argv[2])
```

Output:



Graphical User Interface (GUI)

The GUI version provides a window-based interface where users can type the old and new filenames and rename files by clicking a button. It

improves usability by offering a simple visual interface, making it suitable for general users.

Code:

```
import tkinter as tk

from tkinter import messagebox

import os
```

```
def rename_file():

   old_name = old_filename_entry.get()

   new_name = new_filename_entry.get()
```

```
   if not old_name or not new_name:

       messagebox.showwarning("Warning", "Please enter both filenames.")

       return
```

```
   try:

       os.rename(old_name, new_name)

       messagebox.showinfo("Success", f"File renamed from {old_name} to
{new_name}")

   except FileNotFoundError:

       messagebox.showerror("Error", f"File '{old_name}' not found.")

   except Exception as e:

       messagebox.showerror("Error", f"An error occurred: {e}")
```

```
root = tk.Tk()

root.title("File Renamer")

root.geometry("400x150")   # makes window look professional

root.resizable(False, False)
```

```
tk.Label(root, text="Old Filename:").grid(row=0, column=0, padx=10,
pady=10)

tk.Label(root, text="New Filename:").grid(row=1, column=0, padx=10,
pady=10)
```

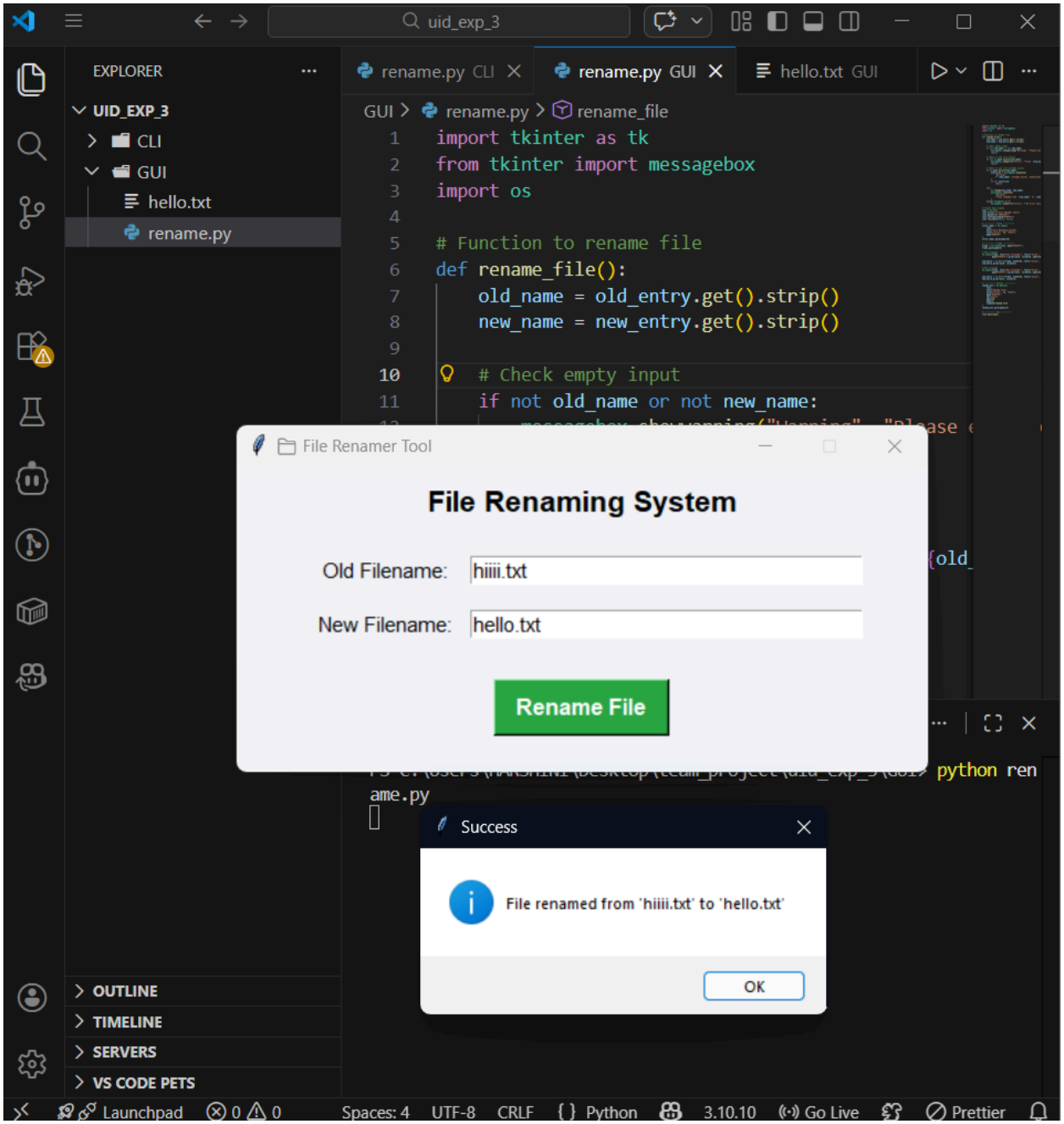
```
old_filename_entry = tk.Entry(root, width=30)

old_filename_entry.grid(row=0, column=1)
```

```
new_filename_entry = tk.Entry(root, width=30)

new_filename_entry.grid(row=1, column=1)
```

```
rename_button = tk.Button(root, text="Rename File", bg="green",
fg="white", command=rename_file)

rename_button.grid(row=2, columnspan=2, pady=15)
```

```
root.mainloop()
```

Output:


uid_exp_3
EXPLORER
UID_EXP_3
CLI
GUI
hello.txt
rename.py
rename.py CLI
rename.py GUI
hello.txt GUI
GUI > rename.py > rename_file
import tkinter as tk
from tkinter import messagebox
import os
# Function to rename file
def rename_file():
old_name = old_entry.get().strip()
new_name = new_entry.get().strip()
# Check empty input
if not old_name or not new_name:
File Renamer Tool
File Renaming System
Old Filename:
hiiii.txt
New Filename:
hello.txt
Rename File
python rename.py
Success
File renamed from 'hiiii.txt' to 'hello.txt'
OK
OUTLINE
TIMELINE
SERVERS
VS CODE PETS
Launchpad
Spaces: 4
UTF-8
CRLF
Python
3.10.10
Go Live
Prettier

Voice User Interface (VUI)

The VUI program allows users to rename files using voice commands. The system listens through a microphone, converts speech into text, and performs the rename operation. This method enables hands-free interaction and demonstrates speech recognition-based control.

Code:

```
import speech_recognition as sr

import os

recognizer = sr.Recognizer()

try:

   with sr.Microphone() as source:

       print("Say: rename old.txt to new.txt")

       recognizer.adjust_for_ambient_noise(source)

       audio = recognizer.listen(source)



   try:

       command = recognizer.recognize_google(audio)

       print("You said:", command)



       words = command.lower().split()

       corrections = {

           "text": "txt",
```

```
        "pdf": "pdf",
        "doc": "doc"
    }
    reconstructed = []
    i = 0
    while i < len(words):
        if i + 2 < len(words) and words[i + 1] == "dot":
            # Merge word + dot + extension
            extension = words[i + 2]
            # Apply corrections to extension
            if extension in corrections:
                extension = corrections[extension]
            reconstructed.append(words[i] + "." + extension)
            i += 3
        else:
            reconstructed.append(words[i])
            i += 1
    old_name = reconstructed[1]
    new_name = reconstructed[3]

    os.rename(old_name, new_name)
    print("File renamed successfully!")
```

```
    except sr.UnknownValueError:
        print("Error: Could not understand audio")
    except sr.RequestError as e:
        print(f"Error: API request failed - {e}")
    except Exception as e:
        print("Error:", e)


except OSError as e:
    print(f"Error: Microphone not available - {e}")
    print("Please ensure a microphone is connected and properly
configured.")
```

Output:

vui.py Welcome

vui.py > ...

```python
                if extension in corrections:
                    extension = corrections[extension]
                reconstructed.append(words[i] + "." + extension)
                i += 3
            else:
                reconstructed.append(words[i])
                i += 1
        old_name = reconstructed[1]
        new_name = reconstructed[3]

        os.rename(old_name, new_name)
        print("File renamed successfully!")

    except sr.UnknownValueError:
        print("Error: Could not understand audio")
    except sr.RequestError as e:
        print(f"Error: API request failed - {e}")
    except Exception as e:
        print("Error:", e)

except OSError as e:
    print(f"Error: Microphone not available - {e}")
    print("Please ensure a microphone is connected and properly configured.")
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\Admin\Desktop\uid> python vui.py
Error: Microphone not available - No Default Input Device Available
Please ensure a microphone is connected and properly configured.
PS C:\Users\Admin\Desktop\uid>
```

```
🎙 Voice File Assistant Started
Say 'rename' to rename a file
Waiting for command...
You said: rename
Say OLD file name (only name)
You said: old
Say NEW file name (only name)
```

```
You said: new
✅ Renamed old.txt → new.txt
```